Project Initialization and Planning Phase

Date	29 September 2026
Team ID	SWTID-2026-8642
Project Title	Import Data Using Transform Maps (Spreadsheet)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to streamline and automate bulk data ingestion into target databases using Transform Maps and Spreadsheet parsing. It tackles manual data mapping inefficiencies, promising reduced duplicate records, faster batch processing, and accurate data synchronization. Key features include automated field mapping, coalesce matching logic, and real-time execution log tracking.

Project Overview	
Objective	The primary objective is to automate the bulk data import process from Excel/CSV spreadsheets into target database tables using Transform Maps, ensuring error-free mapping and data consistency.
Scope	The project comprehensively builds an import staging pipeline, incorporating automated column-to-field mapping, script transformations, duplicate prevention, and exception log tracking.
Problem Statement	
Description	Addressing high error rates, duplicate creation, and time-consuming manual field mapping in traditional spreadsheet upload processes, which adversely affect data integrity.
Impact	Solving these issues will result in improved operational efficiency, seamless database updates, reduced manual intervention, and reliable data governance across systems.

Proposed Solution	
Approach	Employing an Import Set staging mechanism combined with Transform Map logic to parse source spreadsheets, match schema fields automatically, and enforce coalesce rules before updating target tables.
Key Features	• Implementation of automated column-header mapping and transformation scripts. • Real-time coalesce (matching key) execution to detect and prevent duplicate records. • Interactive log tracking dashboard for viewing skipped rows and error reports.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel i5/i7 or equivalent multi-core processor
Memory	RAM specifications	8 GB / 16 GB RAM
Storage	Disk space for data, models, and logs	256 GB SSD / 500 GB Storage
Software		
Frameworks	Web / Backend frameworks	Django / FastAPI / Node.js
Libraries	Spreadsheet parsing and mapping libraries	pandas, openpyxl, xlsxwriter, numpy
Development Environment	IDE / Code Editors	VS Code / PyCharm / Jupyter Notebook

Data		
Data	Source, size, format	User Excel (.xlsx) / CSV files (10,000+ rows sample datasets)